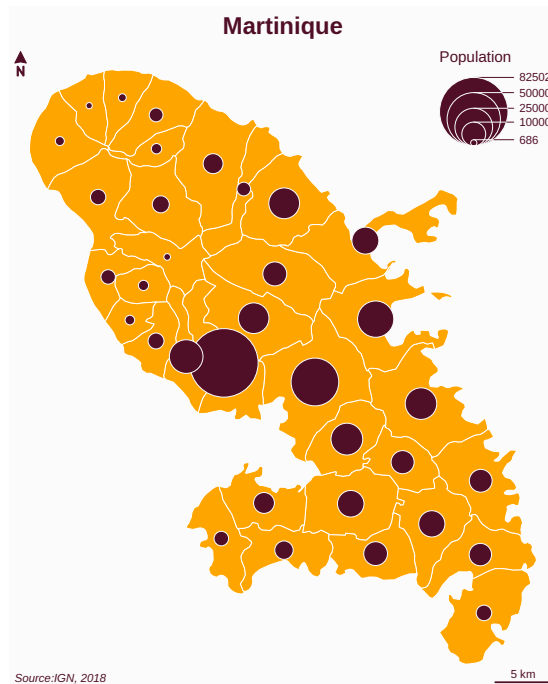


# Thematic maps with mapsf :: CHEAT SHEET



## Base map

```
library(mapsf)
# import the sample dataset
mtq <- mf_get_mtg()
# display a base map
mf_map(mtg, col = "orange", border = "white")
# display symbology
mf_map(mtg, var = "POP", type = "prop",
       leg_title = "Population", add = TRUE)
# map layout
mf_layout(title = "Martinique",
          credits = "Source:IGN, 2018")
```



## Colors

mapsf can use color palettes from `hcl.colors()`  
`mf_get_pal()` is useful to create well-balanced  
asymmetric diverging palettes

```
mf_get_pal(n = c(7, 2), pal = c("Burg", "Mint"))
```



## Symbology

x should be an `sf` object of polygons, lines, or points.

**Choropleth (ratios)**  
`mf_map(x = mtq, var = "var", type = "choro",  
 breaks = "quantile")`

**Typology (categories)**  
`mf_map(x = mtq, var = "var", type = "typo")`

**Proportional Symbols (stocks)**  
`mf_map(x = mtq, var = "var", type = "prop",  
 inches = .3, symbol = "circle")`

**Graduated Symbols (stocks)**  
`mf_map(x = mtq, var = "var", type = "grad",  
 pch = 24)`

**Symbols (categories)**  
`mf_map(x = mtq, var = "var", type = "symb",  
 pch = 21:23)`

**Choropleth Proportional Symbols (stocks - ratios)**  
`mf_map(x = mtq, var = c("var1", "var2"),  
 type = "prop_choro")`

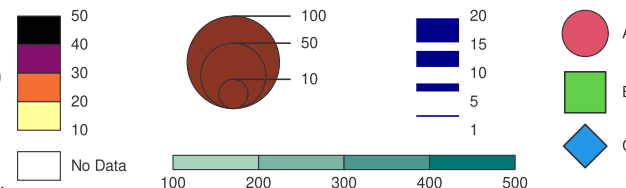
**Colorized Proportional Symbols (stocks - categ.)**  
`mf_map(x = mtq, var = c("var1", "var2"),  
 type = "prop_typo")`

**Choropleth Symbols (stocks - categ.)**  
`mf_map(x = mtq, var = c("var1", "var2"),  
 type = "symb_choro")`

**Raster (single/multi band, continuous, classes, intervals)**  
`mf_raster(x = raster, type = "continuous")`

## Legends

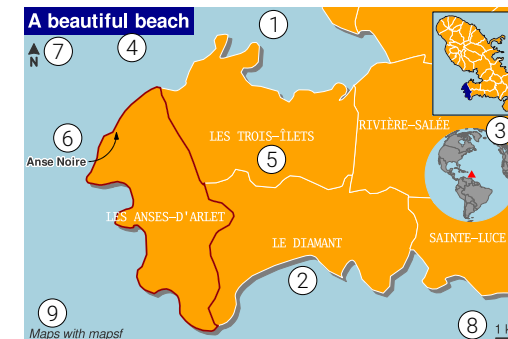
Default legends are plotted with each map.



One can use `mf_legend()` or the `maplegend` package for more customization options.

## Map layout

Customization of the layout design



- ① Set a map theme  
`mf_theme(background = "#aad3df",  
 mar = c(0, 0, 0, 0))`
- ② Create a shadow effect  
`mf_shadow( ... )`
- ③ Create a map inset  
`mf_inset_on(mtg)`  
`mf_map(mtg)`  
`mf_inset_off()`  
Use `mf_inset_on(x = "worldmap")`  
and `mf_worldmap()` to display a globe
- ④ Display a title  
`mf_title( ... )`

## More ?

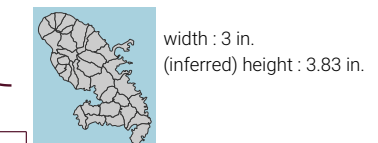
Vignettes on the website:

- > How to use themes
- > How to export maps
- > How to create inset maps
- > How to create faceted maps
- > how to use a custom font family

- ⑤ Display labels  
`mf_label( ... )`
- ⑥ Display an annotation  
`mf_text( ... )`
- ⑦ North arrow  
`mf_arrow( ... )`
- ⑧ Scale bar  
`mf_scale( ... )`
- ⑨ Credits  
`mf_credits( ... )`

## Export maps

`mf_png()` and `mf_svg()`  
export maps in PNG and SVG  
formats. The width/height ratio  
of the map matches the one of  
a spatial object.  
`mf_svg(x = mtq, width = 3,  
 filename = "map.svg")`  
`mf_map(mtg)`  
`dev.off()`



## Class intervals

`mf_get_breaks()` uses `classInt`  
and additional methods (e.g., "Q6", "geom",  
"arith", "msd" or "ckmeans") to classify  
continuous variables.

`mf_distr()` can help to select  
a method for creating class intervals

